

Data Description

The file *newspapers.txt* contains data from the Audit Bureau of Circulations on the circulation of newspapers by county. The variables are:

membernumber	Newspaper's ID number
newspaper	Newspaper's name
stcntyfp	County's ID number (FIPS code)
cnty	County's name
state	State postal code
dailycirc	Circulation of newspaper in this county

One row in the data corresponds to a newspaper-county pair. For example, the row:

membernumber	newspaper	stcntyfp	cnty	state	dailycirc
100050	THE STAR, ANNISTON	1015	Calhoun County	AL	19389

indicates that 19,389 copies of the Anniston Star were sold in Calhoun County, AL. Note that the county fips code (stcntyfp) uniquely identifies a county, while the county name (cnty) does not.

Objective

The goal of the exercise is to write a short script that produces a dataset at the *county-pair* level that identifies the total circulation in one county of newspapers headquartered in another county. A newspaper is defined to be headquartered in the county where its circulation is highest. The variables in output dataset should be:

stcntyfp_1	County 1's ID number
cnty_1	County 1's name
state_1	County 1's state postal code
stcntyfp_2	County 2's ID number
cnty_2	County 2's name
state_2	County 2's state postal code
dailycirc	Total circulation of newspapers from county 1 in county2

County 1 is the headquarter county. County 2 is the county in which we measure circulation. The dataset should include one observation for each pair of counties for which the circulation of county 1 in county 2 is positive, including cases where County 1 is the same as County 2. The final dataset should be sorted by stcntyfp_1 then stcntyfp_2 and if it is created in a statistical program it should have appropriate variable labels. You should include observations for a newspaper's circulation in its home county (i.e., where stcntyfp_1 equals stcntyfp_2).

In evaluating the project we will not only look at whether the output file is correct, but also at: (i) whether the algorithm is efficient and concise, (ii) whether the code is clean, easy-to-read, and well-commented, and (iii) how quickly the code was produced.

So we can evaluate (iii), we ask that you *accurately record the total time spent on the project beginning when you first read through the instructions.*